

DRAGOMAN, BulgariaWMO: **156050**Lat: **42.933N**Lon: **22.933E**Elev: **2349**StdP: **13.49**Time Zone: **2.00 (EUE)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	12.5	3.6	7.2	10.6	7.7	8.9	14.5	24.9	30.3	22.0	26.7	4.5	90	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	87.7	68.0	84.4	67.0	81.3	65.7	69.9	82.7	68.3	81.3	66.8	79.1	6.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	104.3	75.3	64.0	97.3	73.5	62.4	91.9	71.9	35.4	83.3	33.9	81.6	32.7	79.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.6	16.6	-2.4	93.3	6.3	3.0	-6.9	95.4	-10.6	97.2	-14.2	98.9	-18.8	101.1		
(5)			-2.7	73.5	6.1	2.2	-7.1	75.1	-10.7	76.3	-14.1	77.6	-18.5	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	29.4	31.3	39.5	47.8	57.2	63.2	67.4	67.3	59.5	50.0	40.7	31.2
	DBStd	15.47	8.73	8.90	8.30	7.05	6.11	5.89	5.18	5.22	6.53	7.76	8.06	8.88
	HDD50	2604	639	523	333	115	12	1	0	0	6	99	294	582
	HDD65	6200	1104	943	790	516	247	102	33	34	188	467	730	1046
	CDD50	2170	0	0	8	50	236	396	541	536	290	98	14	1
	CDD65	290	0	0	0	0	6	46	109	105	22	2	0	0
	CDH74	3285	0	0	0	14	111	544	1150	1146	300	20	0	0
	CDH80	938	0	0	0	1	11	116	365	366	76	3	0	0
Wind	WSAvg	7.1	7.2	8.2	8.5	8.1	7.2	6.1	6.3	6.1	6.5	6.6	6.9	7.4
Precipitation	PrecAvg	24.0	1.5	1.5	1.5	2.1	2.8	2.8	2.1	2.1	2.0	2.1	1.6	1.8
	PrecMax	34.8	4.0	2.8	2.8	3.9	6.3	5.8	4.3	6.8	6.5	4.5	4.8	3.7
	PrecMin	14.5	0.3	0.3	0.2	0.4	0.8	0.8	0.4	0.3	0.3	0.2	0.0	0.4
	PrecStd	5.1	1.0	0.7	0.7	0.9	1.2	1.3	1.1	1.7	1.4	1.3	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	59.1	67.5	75.5	81.4	87.1	93.2	92.1	86.5	76.3	67.2	56.2
		MCWB	46.1	49.3	54.4	59.7	63.3	69.0	67.9	68.8	66.7	61.7	56.0	50.3
	2%	DB	48.5	52.9	62.5	67.8	76.9	83.4	87.9	87.7	81.6	71.0	61.7	51.5
		MCWB	42.1	46.1	50.5	54.8	61.9	68.2	67.9	68.0	64.3	58.2	52.6	46.2
	5%	DB	44.8	48.7	58.1	63.8	73.7	80.1	84.2	84.2	76.9	66.9	56.9	47.9
		MCWB	41.0	43.5	48.3	52.1	61.0	66.6	67.1	66.6	61.8	56.2	49.9	44.3
	10%	DB	41.2	44.5	53.7	60.4	70.5	76.7	80.6	80.5	72.6	63.0	53.0	44.3
		MCWB	38.2	40.4	45.9	50.6	59.5	65.0	65.8	64.9	59.9	54.6	47.5	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	51.3	55.9	60.5	65.6	71.9	72.3	71.5	67.4	63.2	56.8	50.7
		MCDB	53.0	55.7	65.2	74.4	77.1	81.0	85.4	84.4	84.0	74.0	65.4	55.4
	2%	WB	43.4	46.7	51.7	56.2	63.6	69.6	69.6	69.2	65.1	59.5	53.4	47.0
		MCDB	47.0	52.3	61.2	65.1	73.8	80.5	82.7	84.0	79.2	68.1	60.0	50.5
	5%	WB	41.0	43.7	48.9	53.6	61.7	67.7	68.1	67.6	63.0	57.4	50.8	44.5
		MCDB	44.6	48.4	56.8	61.7	71.5	77.9	81.1	81.9	74.6	64.5	55.8	47.6
	10%	WB	38.4	40.7	46.5	51.5	59.9	65.8	66.5	65.9	60.9	55.4	48.1	41.8
		MCDB	41.1	44.1	53.1	58.9	69.4	75.6	78.5	78.4	70.6	61.5	52.2	44.2
Mean Daily Temperature Range	5% DB	MDBR	12.4	14.0	17.3	18.3	19.4	20.3	22.1	22.5	20.6	18.3	15.9	11.6
		MCDBR	18.6	22.9	27.0	26.1	25.8	27.2	28.8	30.1	29.2	25.5	23.8	16.5
		MCWBR	13.5	17.0	17.2	14.3	13.8	13.9	12.5	13.0	14.4	14.5	15.4	12.2
	5% WB	MCDBR	17.9	22.0	25.1	22.9	23.7	25.4	25.4	26.0	26.3	22.3	20.4	14.5
		MCWBR	13.3	16.9	16.9	13.6	13.4	14.0	12.3	12.2	13.6	13.4	13.7	11.1
Clear-Sky Solar Irradiance	taub	0.312	0.341	0.384	0.442	0.430	0.422	0.427	0.440	0.413	0.372	0.338	0.313	
	taud	2.346	2.281	2.220	2.069	2.168	2.229	2.216	2.179	2.232	2.339	2.399	2.386	
	Ebn at Noon	259	269	270	261	268	270	267	259	257	254	247	247	
	Edh at Noon	26	33	40	50	46	44	44	44	39	31	25	23	
All-Sky Solar Radiation	RadAvg	490	757	1086	1411	1701	1937	1998	1796	1308	899	560	415	
	RadStd	73	102	129	171	152	167	146	156	109	118	63	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		+0.94	N/A	N/A	N/A	+0.67	N/A	N/A	-265	N/A	N/A

Nomenclature: See separate page